STUDENT ID NO

MULTIMEDIA UNIVERSITY

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FINAL EXAMINATION

TRIMESTER OCTOBER/NOVEMBER 2025 (TERM ID: 2530)

CMT1114 – MATHEMATICS I
(Foundation in Computing)

21 FEBRUARY 2026
9.00 – 11.00am
(2 Hours)

INSTRUCTIONS TO STUDENT

1. This question paper consists of three pages with **FIVE** questions.
2. Attempt **ALL** questions. All questions carry equal marks and the distribution of the marks for each question is given.
3. Please write all your answers in the answer booklet provided.
4. **You are required to write proper steps to obtain maximum marks.**

Question 1 [10 marks]

a) Simplify the expression and write your final expression as a fraction with no negative exponents.

$$\left(\frac{x^4y^{-3}}{2}\right)^{-2}\left(\frac{x^3y}{8}\right)$$ (2 marks)

b) Rationalize the denominator for $\frac{5}{4-2\sqrt{3}}$ and simplify it. (2 marks)

c) Factorize the polynomial $8x^3-27$. (2 marks)

d) Simplify the following expression and write your final answer as a single fraction.

$$\frac{5}{(x+2)(x-1)}-\frac{1}{(x+2)(x+1)}$$ (2 marks)

e) Express $\frac{2-5i}{1+3i}$ in the form $a+ib$ where a and b are real numbers.

(2 marks)

Question 2 [10 marks]

a) Solve the equation $|3x-2|=16$. (2 marks)

b)

i) Solve the inequality $\frac{x+3}{(x-1)(x+2)}<0$. Clearly show your Sign Diagram and write your final answer in interval notation.

ii) Find the domain of the function $g(x)=\sqrt{\frac{x+3}{(x-1)(x+2)}}$. Write your final answer in interval notation.

(5 marks)

c) Solve $\sqrt{5x+1}=x+1$. Remember to check your answer(s).

(3 marks)

Continued ...

Question 3 [10 marks]

a) Given two functions $f(x)=\dfrac{1}{x+1}$. and $g(x)=\dfrac{x+2}{x+1}$.

i) Find the domain of g. Give your answer in interval notation.

ii) Find $(f \circ g)(3)$.

iii) Find $f^{-1}(x)$. (5 marks)

b) Given a polynomial function $f(x)=2(x+1)^2(x-1)(x+3)$.

i) What is the degree of f?

ii) Find the zeros of f and their multiplicities.
At each zero, determine whether the graph of f crosses or touches the x-axis.

iii) Find the y-intercept of the graph of f.

iv) Determine the end behavior of f. (5 marks)

Question 4 [10 marks]

a) The graph of $f(x)=3^{2x+h}+4$ passes through point $(2,13)$. Find the value of h.

Hence, find the y-intercept of the graph.

(2 marks)

b) Solve the following equations:

i) $\dfrac{125}{25^y}=5^{1-3y}$

ii) $\log_2\left(x^2+2\right)=1+\log_2(x+5)$

(5 marks)

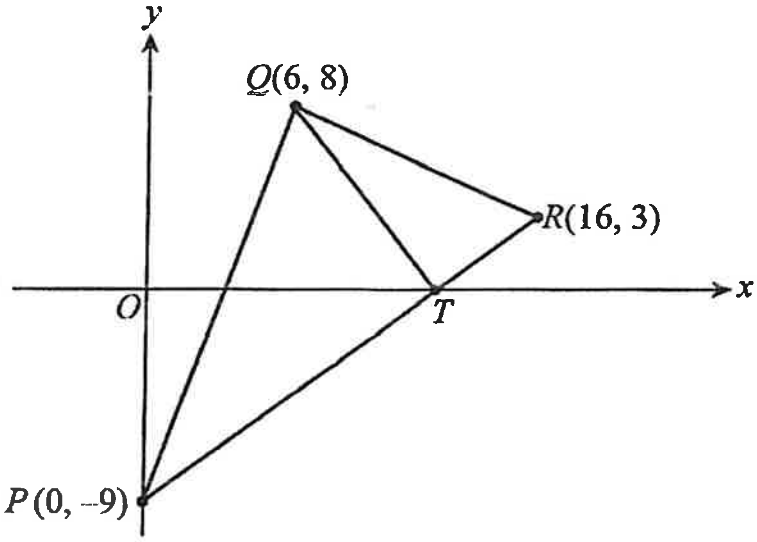
c) Use **long division** to find the quotient and remainder when the polynomial function $P(x)=2x^3+x^2-3x+1$ is divided by (x^2-3).

(3 marks)

Continued ...

Question 5 [10 marks]

a) Diagram below shows a triangle *PQR*. Point T lies on the x-axis. It is given that straight lines *PR* and *QT* are perpendicular.



i) Find the gradient of PR and QT.
ii) Find the equation of the straight line QT.
iii) Find the coordinates of T.
iv) If the straight line QT is extended to a point S such that $\frac{QT}{TS} = \frac{2}{3}$, find the coordinate of S.

(7 marks)

b) Given an equation of a circle $x^2 + y^2 + 6x - 4y - 12 = 0$.
i) Express the equation in the form $(x-h)^2 + (y-k)^2 = r^2$ where *h*, *k* and *r* are constants.
ii) Find the radius and centre of the circle.

(3 marks)

End of Page.